

Tensegrity Tower

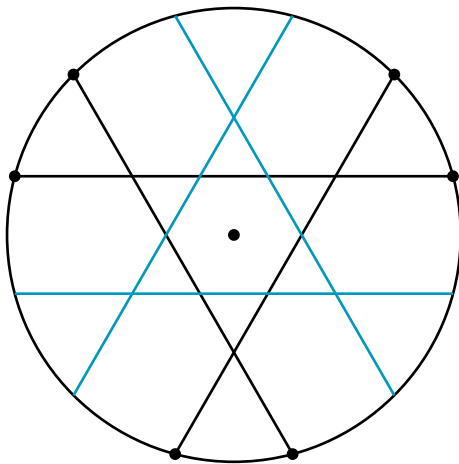
Following the pattern of the [Needle Point Tower](#) at the Hirshhorn Museum, except without the extra large bottom layer.

Note: all angles are measured in turns.

Layout

Looking from above, each layer of sticks makes a triangle with bits sticking out. Each layer is a mirror of the layer below (it twists in the opposite direction). According to the thesis [A Practical Guide to Tensegrity Design](#), the unique correct rotation of the tops of the sticks relative to the bottoms of the sticks (the change of θ in polar coords) is $5/12$.

Here's the view from above of one layer in black, and the layer above in blue:



Degrees of Freedom

A completed tower must have exactly 6 degrees of freedom (3 translational and 3 rotational). Each stick starts with 5 degrees of freedom (3 translational and 2 rotational). There are three sticks per layer. Each string eliminates one degree of freedom of motion, so if m is the total number of strings, then $5 * 3n - m = 6$, so $m = 15n - 6$.

Call the number of string-ends m' . There are two string-ends per string, so $m' = 2m$.

Call the number of stick-ends p . There are two stick-ends per stick and three sticks per layer, so $p = 6n$.

Putting these together, we have $m' = 2m = 30n - 12 = 30\left(\frac{p}{6}\right) - 12 = 5p - 12$. So there

are 5 strings coming out of each end of a stick, except for the bottom-most and top-most layers which have 4 strings per stick-end.

Definitions

- n is the number of layers (so there are $3n$ sticks).
- $m = 15n - 6$ is the total number of strings.
- $r = 1$ is the radius of the cylinder circumscribing the tower.
- s is the length of each stick.
- h is the height from the bottom of each stick to the top.
- d is the dip from one layer of the tower to the next (that is, the change in height from the bottom of a layer to the top of the layer below it).

Measurements

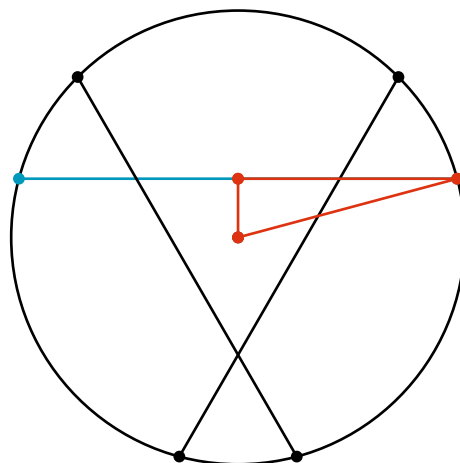
There are sticks, plus four kinds of strings:

- *Sticks* are grouped in layers, three sticks per layer.
- *Triangular Strings* form a triangle at the very top and bottom of the tower.
- *Hexagonal Strings* form wobbly hexagons at the intersections between layers.
- *Inter-Layer Spinal Strings* go from the bottom of a stick in a layer to the top of a different stick in the same layer.
- *Intra-Layer Spinal Strings* go between sticks of different layers.

See the helpful diagram on page 22 of [Tensegrity and Weaving](#).

Sticks

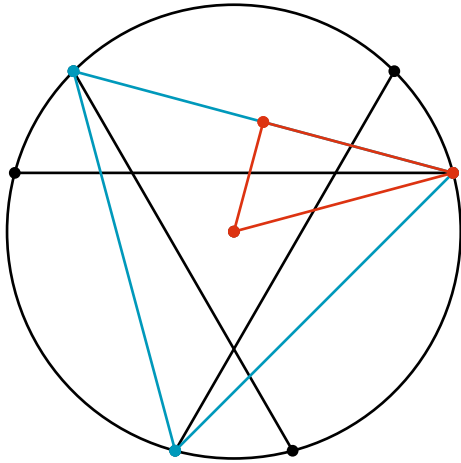
Let's measure the length of one of the sticks.



This triangle has an angle of $\frac{2.5}{12}$ turns, so the length of the blue line when viewed from above is $2 \sin\left(\frac{5}{24}\right) = \frac{1+\sqrt{3}}{\sqrt{2}}$. The change in height is h , giving a total length of $\sqrt{\left(\frac{1+\sqrt{3}}{\sqrt{2}}\right)^2 + h^2} = \sqrt{2 + \sqrt{3} + h^2}$.

Triangular Strings

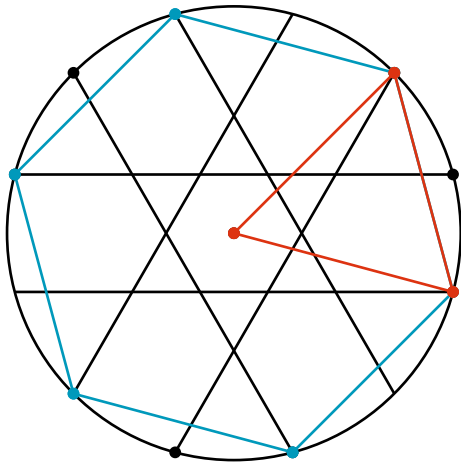
There are strings forming a triangle at the very bottom and very top of the tower:



This is a $\frac{1}{12}, \frac{2}{12}, \frac{3}{12}$ triangle, so each of these strings has length $\sqrt{3}$.

Hexagonal Strings

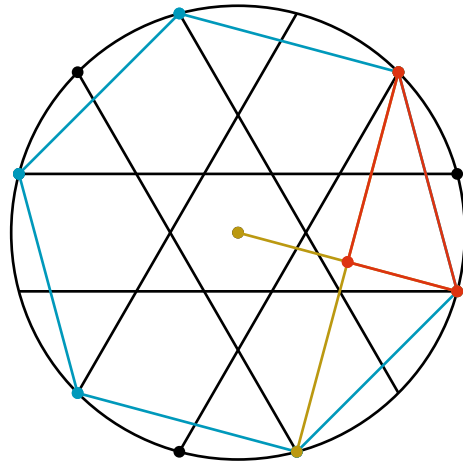
There are strings connecting the top of one layer to the bottom of the next, forming a wobbly horizontal-ish hexagon.



This is an equilateral triangle, so the length of each hexagonal string when viewed from above is 1. The change in height is d , so the total length is $\sqrt{1 + d^2}$.

The choice of d isn't arbitrary, though. I'm not sure how to calculate it precisely, but my best

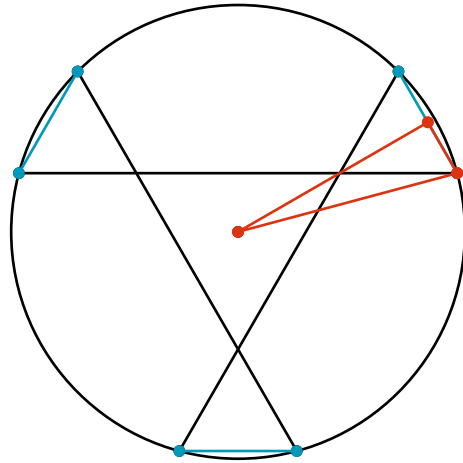
guess is that the horizontal and vertical displacements of the bottom of a stick from the center point of the tops of the two sticks supporting it should be equal.



This is a $\frac{1}{12}, \frac{2}{12}, \frac{3}{12}$ triangle, making my estimate $d = \frac{1}{2}$. Making the hexagonal string length be $\sqrt{1 + d^2} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$.

Intra-Layer Spinal Strings

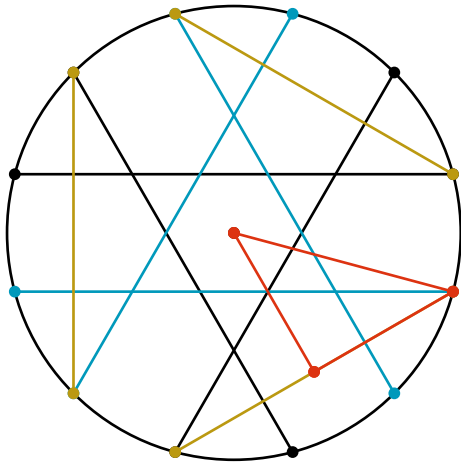
Each layer has three vertical-ish strings going from the bottom of its sticks to the tops of other sticks.



This triangle has angle $1/24$, so the string when viewed from above has length $2 \sin\left(\frac{1}{24}\right) = \frac{\sqrt{3}-1}{\sqrt{2}}$. The change in height is h , so the total length is $\sqrt{2 - \sqrt{3} + h^2}$.

Inter-Layer Spinal Strings

There are also vertical-ish strings going between layers: from the bottom of a stick to the bottoms of the sticks above and below it, and likewise for the tops of sticks.



This is an isosceles right triangle, so the yellow line (viewed from above) has length $\sqrt{2}$. The change in height is $h - d$, so the total length is $\sqrt{2 + (h - d)^2}$.

Summary

(No longer assuming $r = 1$, which scales all measurements up to this point by a factor of r .)

- $s = \sqrt{2 + \sqrt{3} + \left(\frac{h}{r}\right)^2} r$
- $d = \frac{r}{2}$
- Triangular string length = $\sqrt{3}r$
- Hexagonal string length = $\frac{\sqrt{5}}{2}r$
- Intra-layer spinal string length = $\sqrt{2 - \sqrt{3} + \left(\frac{h}{r}\right)^2} r$
- Inter-layer spinal string length = $\sqrt{2 + (h - d)^2} r$.

Picking $h = \sqrt{3}r$:

- $s = \sqrt{5 + \sqrt{3}}r$
- $d = \frac{r}{2}$
- Triangular string length = $\sqrt{3}r$
- Hexagonal string length = $\frac{\sqrt{5}}{2}r$
- Intra-layer spinal string length = $\sqrt{5 - \sqrt{3}}r$
- Inter-layer spinal string length = $\sqrt{2 + \left(\sqrt{3} - \frac{1}{2}\right)^2} r = \sqrt{\frac{21}{4} - \sqrt{3}}r$.

If $s = 1219\text{mm}$:

- $s = 1219\text{mm}$
- $r = 470\text{mm}$
- $h = 814\text{mm}$
- $d = 235\text{mm}$
- Triangular string length = 814mm . Count of 6 for $4.884m$.
- Hexagonal string length = 525mm . Count of $6(n - 1) = 54$ for $28.350m$.

- Intra-layer spinal string length = 849mm . Count of $3n = 30$ for $25.470m$.
- Inter-layer spinal string length = 881mm . Count of $6(n - 1) = 54$ for $47.547m$.

Total string count is 144, which matches the $m = 15n - 6$ formula from the degrees-of-freedom analysis at the start of the document.

Total string length is $106.251m$.

Materials

Sticks 30 x 4' long 3/8" diameter wooden dowels. Weight about 0.4 lb, for a total of 12 lb. Where does one get wood dowels? From wood-dowels.com of course.

Bolts 60 x 3/16" hanger bolts. The ones I prototypes with are 2" long in total, with 1" of screw, 5/8" of 3/16" bolt, and 3/8" blank in the middle. For the rest I'll get [7/8+5/8 inch 10-24 hanger bolts](#).

Nuts 60 x 3/16" nuts. Getting [these](#).

Washers About 300 washers, for either end of the strings. The ones I got for the prototype have a 6mm inner diameter, and are not very wide or thick. These [expensive shims](#) have the perfect dimensions, but might not be strong enough. Here are [cheap washers](#) that are quite large, but cheap and strong.

String Lacking a precise total string count, my eyeball says there are about 5 string-ends per stick-end on the Needle Tower. There are $30 * 2 = 60$ stick-ends, making about $60 * 5 = 300$ string-ends. Two string-ends per string, so 150 strings. Most are roughly 3-feet, giving roughly 450 feet of string. [My previous estimate was 200feet, something is off!] Fortunately we live in the future, so you can get [black 1mm cord that's rated for 350lbs](#).

Paint Paint or wood stain. Maybe deep red paint.

Protectant Something to put over the paint to make it weather resistant.

[McMaster](#) has lots of specific hardware.

Links

- [Tensegrity links](#)
- [Tower diagram](#) (pg 22)
- [Vague tutorial](#)

- [High quality Needle Tower pic](#)
- [Thesis on tensegrity math](#)
- [Short glowy tensegrity tower](#)
- [Needle tower](#)

Open Questions

- Is there one correct choice for d relative to h , or can it be whatever? The failure mode would be a floppy tower because the sticks can settle into a lower-string-length configuration.
- How to tension final cords?